SHIVENDRA PRATAP SINGH

AI/ML Engineer | Generative AI Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer and Computer Science (AI) undergraduate building end-to-end machine learning and Generative AI applications. Work spans multi-agent workflows with LangGraph, RAG pipelines on Groq-hosted LLMs, PyTorch and TensorFlow models, and FastAPI services with SQL storage, covering the path from modeling to Docker and cloud deployment. Deployed 4 live applications on Render and Streamlit Cloud. Solved 150+ LeetCode problems in C++.

EDUCATION

Bachelor of Technology (B.Tech) – Computer Science (AI) | IIMT Group of Colleges | Expected 2028

Senior Secondary Education (CBSE) | Sukhrawati Public School | 2024

TECHNICAL SKILLS

- **Languages:** Python, C++, SQL, C
- **Generative AI & Agents:** LLMs, Transformers, Agentic AI, Multi-Agent Systems, LangGraph, LangChain, RAG, Prompt Engineering, Embeddings & Vector Retrieval, Groq API
- **ML, DL & NLP:** PyTorch, TensorFlow/Keras, Scikit-learn, XGBoost, NLP, Transfer Learning, Hyperparameter Tuning (Optuna, Keras Tuner), Explainable AI
- **Backend & Tools:** FastAPI, REST APIs, SQLite, SQLAlchemy, Pydantic, Pandas, NumPy, Streamlit, Docker, Git, GitHub
- **CS Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS

PROJECTS

EduPath 2.0 — *Personalized Learning & Skill Gap Agent* | *LangGraph, Groq LLM, SQLite, Pandas, Plotly* Live Demo | GitHub

- Built an AI-powered personalized learning platform that analyzes learner profiles and identifies skill gaps based on target career roles.
- Developed an evidence-based competency mapping system with skill prioritization, prerequisite analysis and personalized learning roadmaps.
- Implemented learning objectives, recommended resources, practice tasks, interview preparation and progress tracking.
- Integrated interactive Plotly charts to visualize skill gaps, learning progress, evidence coverage and learner readiness.

EduPilot — *Adaptive AI Learning Copilot* | *FastAPI, RAG, Groq LLM, SQLAlchemy, Docker* Live Demo | GitHub

- Developed a RAG pipeline (extraction, chunking, embeddings, retrieval) over 2 sources, PDF notes and YouTube transcripts, to ground answers and reduce hallucination in English, Hindi and Hinglish.
- Designed topic-level mastery scoring with 3 tiers (Strong ≥75%, Revise 50–74%, Weak <50%) that drives quiz generation, weak-topic detection and study-plan recommendations.
- Architected a decoupled system: FastAPI backend with 5 REST route modules and SQLAlchemy/SQLite storage, containerized with Docker on Render, plus a Streamlit UI.

Riskify — *AI Student Early-Warning System* | *PyTorch, Scikit-learn, Groq LLM* Live Demo | GitHub

- Trained a PyTorch neural network to predict academic risk from 6 signals (attendance, quizzes, assignments, LMS activity, study hours, backlog), reaching ~0.86 validation AUC on synthetic data.
- Added gradient-based explainability and a Groq LLM action-plan coach with a rule-based fallback.

Credit Card Fraud Detection — *Artificial Neural Network* | *TensorFlow/Keras, Scikit-learn* Live Demo | GitHub

- Built a Keras ANN on 30 input features for imbalanced fraud detection using Keras Tuner, Dropout, L2 regularization, early stopping and threshold tuning; scored 96.6% ROC-AUC, 87% precision, 76% recall.
- Launched a Streamlit app with single and batch CSV prediction, input validation and downloadable reports.

ACHIEVEMENTS

- Participated in Agentic AI Hackathon (Product Space): built EduPath, a 7-agent LangGraph system, deployed live.
- Solved 150+ problems on LeetCode in C++ across data structures and algorithms.